

Technology





About the Course

This course will guide students in developing meaningful skills using modern tools. Students will work hands-on with various technologies, from digital media creation to coding. Throughout the course, they will explore foundational principles and use what they have learned to solve problems and create original media while reflecting on the implications of technology and considering questions about privacy, the digital divide, and the role of innovation in society. This course will equip students to think critically about the world of technology and its impact on their lives and future.

NOTE: Students need a Google email account, but no books are required for this course.



Placement & Combining Tips

May be suitable for Grades 7-8.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Technology 2 times/week 30 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Technology				
Technology		Technology		



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Technology

You will need a Google account before starting this course. The student's experience will be greatly enhanced if they have access to a mobile device to test out the apps they will be making.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
(No Books Assigned)



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ [View Basic Supplies](#)
(No Subject Supplies Assigned)



Quick Links

Technology

- ∞ [Extra Helpings](#)
- ∞ [Course Overview](#)
- ∞ [Term 1 Exam Link](#)
- ∞ [Term 3 Exam Link](#)

Click THIS text or scan the QR code for links.



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How To Approach



Prepare

Preview the lesson. Make sure you have the required materials (paper and writing utensil) ready and links (Learning Journal, HTML Word Journal) open before beginning so they are easily accessible during the lesson.



Skim

Begin each day by briefly skimming the whole lesson. Consider the THINK prompt at the start of the lesson to activate curiosity and access prior knowledge. Consider how today's skills build on what you've previously explored.



Watch/Read

Watch the assigned videos or read the provided articles, pausing often to take note of interesting or important ideas. Good notes capture both facts and your connections or questions.



Practice

After exposure to new content, follow along, experiment, and play with the lesson concept, topic, or idea. This will include adding to and editing your HTML Word Journal, learning journals, and Technology Timeline. This could also include designing a logo or trying coding exercises. Try thoughtful things. Allow yourself to make mistakes. Observe what happens and what, in happening, causes something else to happen. Record your findings.



Reflect

Reflect in writing on what you learned, noticed, or found challenging during the lesson. Update projects (e.g., inserting images, improving layouts) to solidify new skills.



Share

When possible, share what you create during the lesson with a teacher or facilitator, friends, and family members. This can help highlight the real-world relevance of skills like web design, historical inquiry, and creative problem-solving.



Term 1

WEEK 1 30m Technology - Lesson 1

Introduction to this Course

Materials: Google Docs Learning Journal (you will create this today)

→ INTRO

You will create a basic website or blog using Google Docs to access your prior knowledge and reflect on, record, and share what you are learning throughout this course.

→ VIEW

∞ Link: Course Overview

→ OPEN

- ∞ Link: My Learning Journal Template
- Click the blue "Use Template" button.
- Send the URL (link or website address) to your educator so they can see the updates you make as you go through the lessons.

You will use this as a learning journal throughout this first term, and you will respond within the journal you just created. Many lessons will begin and end with an opportunity to reflect on what you will learn or what was covered. This is your journal, and we encourage you to make it beautiful as you learn more about design throughout this course. Do not share any personal information. This will be referred to in future lessons as your "Google Docs Learning Journal."

→ REFLECT

- Use the landing page (opening page) of your Google Docs Learning Journal to respond to the prompts under "What Stands Out to You About the Course Overview."
- Read the section "Goals for Term 1." Begin thinking about your philosophy of education by comparing the two statements that are provided and answer #3.

★ TEACHER TIP

Watch these videos if you are unfamiliar with Google Docs.

∞ Video Link: Using Google Docs to Create Web Pages

∞ Video Link: Dive deep into Using Google Docs

★ TEACHER NOTE

When you see instructions using this symbol ">", you are being given steps in order to navigate within the webpage or program/app.

WEEK 1 30m Technology - Lesson 2

Introduction to HTML #1

Materials: your Google Docs Learning Journal

→ INTRO

You will learn about the invention of HTML and its creator, Sir Tim Berners-Lee.

→ OPEN

Your Google Docs Learning Journal and scroll down to the first lesson section called "Intro to HTML: Sir Tim Berners-Lee."

→ THINK

Spend 5 minutes answering any of the following questions within your journal: (Anytime you do not know how to answer a question, write that! Starting by recognizing and recording what you do not know increases the possibility that you will feel good about learning and that the idea will stick in your mind.)

- What is the World Wide Web, and who invented it?
- What is the difference between the World Wide Web and the Internet?

→ READ OR WATCH

∞ Article Link: Sir Tim Berners-Lee (read)

∞ Video Link: Sir Tim Berners-Lee (video) (15 mins)

You will be expected to write a narration of this video, so pause the video



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and jot down notes when you notice something noteworthy.

→ REFLECT

Complete your first entry in your Google Docs Learning Journal by spending 10 minutes writing a brief narration summarizing Berners-Lee's journey from building devices for model trains to inventing HTML.

WEEK 2 30m Technology - Lesson 3

Introduction to HTML #2: Create a Technology Word Journal using HTML

 Materials: your Google Docs Learning Journal, notepad on Windows or TextEdit on Mac, and Technology Word Journal

→ OPEN

Open your Google Docs Learning Journal and scroll down to the second lesson section called "Intro to HTML."

→ THINK

Spend 5 minutes answering the following question:

- How do websites know how to show words, pictures, and videos in the right places?

→ WATCH (14 MIN)

∞ Video Link: Video #00

∞ Video Link: Video #01

→ OPEN

In this section, you will either open NOTEPAD or TEXTEDIT, depending on which kind of computer you are using.

∞ Link: Alveary Technology 10 Course Word Journal

- Go to File> download> Plain Text (.txt) > open it in Notepad (PC) or TextEdit (Mac)

→ FOLLOW DIRECTIONS

- Follow these directions for using NOTEPAD on a PC:
 - Go to File> Save As > and save a working copy of the .txt file as backupwordjournal.txt > Save it another time with the extension .HTML instead of .txt.
- Follow these directions if you are using TEXTEDIT on a MAC:
 - Hold down the Option key while choosing File, then Save As > type the name of the copy as backupwordjournal.txt and choose where to save it > save > Save it another time, holding down the Option key while choosing File, then Save As> type the name of the copy as backupwordjournal.HTML and choose where to save it > save > use .HTML.

→ FOLLOW DIRECTIONS FOR PC AND MAC

- Now, close the copy you created and reopen it. It will open in your browser, and you will see that it has been rendered with the HTML formatting.
- Open the .txt document and examine how the HTML tags impacted the resulting webpage. Don't fret if you mess it up! Remember, you saved a backup.

→ REFLECT

You will notice that entries for the terms "Technology" and "Computer" have been completed. Using these as an example, add an entry for one of the following terms to the new HTML Word Journal you just created in the Notebook or TextEdit program.

- HTML, HTML Tag, HTML element, or Markup Language



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WEEK 2 30m Technology - Lesson 4

Introduction to HTML #3

Materials: your Google Docs Learning Journal and HTML Word Journal.

→ INTRO

You will consider some given reasons for learning HTML and interact with HTML tags while receiving instant feedback.

→ OPEN

Open your Google Docs Learning Journal and navigate to the next available lesson section.

→ THINK

Spend 5 minutes answering any of the following questions:

- Have you ever right-clicked on a webpage and selected "View Page Source" or "Inspect"? Try it! What do you see?
- Why do you think all websites follow some basic structure rules to display content?
- Write about a website that surprised you by how "new" and "different" it looked.

→ VIEW AND PRACTICE

Article Link: HTML Introduction by Geeks for Geeks

- Read the introduction.
- Scroll until you find the embedded "Basic HTML Code Example" and click on the triangle play button.
- Observe the output.
- Just for fun, insert "<marquee>" and "</marquee>" tags inside the opening header tag "<h1>" and the closing header tag "</h1>" as in the following example:
<h1><marquee>Welcome to My Webpage</marquee></h1>
- Click on the triangle play button and observe the output.

→ REFLECT

While referring to the provided Geeks for Geeks HTML Introduction, spend 5-10 minutes completing your Google Docs Learning Journal answering the following questions:

- Of the five provided, which is the best reason for learning HTML?
- What does the "marquee" tag do?

→ OPEN

Your working copy of the Word Journal.txt

- Create entries for 2 more of the following terms: HTML, HTML Element, HTML Tag, or Markup Language.
- If you are happy with the changes you have made to your HTML Word Journal so far, this might be a good time to consider updating the backup.

• TERMINOLOGY

Article Link: A Simple Guide to HTML

★ TEACHER TIP

Don't fret if you mess it up! Remember, you saved a backup.

WEEK 3 30m Technology - Lesson 5

What are Computers?

Materials: your Google Docs Learning Journal and HTML Word Journal

→ OPEN

Open your Google Docs Learning Journal and navigate to the next lesson section.

→ THINK

Spend 5 minutes answering the following questions about these terms: hardware, operating system, servers, monitor, video processor, LCD



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displays, LED displays, processing, ports, USB, mother board, CPU, heat sync, RAM, hard drive, SSD, expansion card, applications, surge protector, ergonomics, desktop, task bar, menu bar, doc, modem, router, ethernet, Wi-Fi, the cloud, malware, virus, spyware, trojan horses, ransomware, and keyloggers.

- Which of the following terms do you know well?
- Which ones do you want to know more about?

→ VIEW AND NARRATE

∞ Video Link: Computers Playlist

You will be expected to write about what you watch in these videos, so pause the video and jot down notes when you notice something noteworthy.

- Spend 10 minutes watching the videos that touch on the terms you wanted to know more about.

→ OPEN

Your working copy of your HTML Word Journal.txt

- Create at least 2 entries in your HTML Word Journal from the terms you explored today.

→ REFLECT

Reflect on what the computer terms you already knew before starting this lesson had in common. In your Google Docs Learning Journal, compare those terms to the terms you looked up. What area or areas of computing are you less familiar with?

WEEK 3 30m Technology - Lesson 6

What is Computer Science?

📁 Materials: your Google Docs Learning Journal and HTML Word Journal

→ OPEN

Open your Google Docs Learning Journal and navigate to the next available lesson section.

→ THINK

Briefly answer the following questions about the following terms: software, binary code, algorithm, sequences, selections, loops, programming languages, JavaScript, Python, Java, C++, HTML, and CSS.

- Which of the following terms do you know well?
- Which ones do you want to know more about?

→ WATCH

∞ Video Link: Computer Science Basics Playlist (15 mins)

You will be expected to write about what you watch in this course, so pause the video and jot down notes when you notice something noteworthy.

- Create at least 3 entries in your HTML Word Journal from the terms you explored today.

→ REFLECT

Complete your Google Docs Learning Journal entry for this lesson by reflecting on what the computer science terms you already knew well enough before starting this lesson had in common. Compare those terms to the terms you looked up; what area or areas of computing are you less familiar with?